Master Theorem

1) $a > 0$
2) $b > 1$ } must be constant
3) $T(n) = aT\left(\frac{n}{b}\right) + f(n)$

4) $f(n)$ musta be +ve

5) watershed function $= n^{\log_b a}$

1) $WS > f(n)$ then apply Case 1
2) $WS = f(n)$ then apply Case 2
3) $WS < f(n)$ then apply Case 3

Case 1

$\epsilon > 0$

$f(n) = \Theta\left(n^{(\log_b a) - \epsilon}\right)$

$T(n) = \Theta\left(n^{\log_b a}\right)$ ← (WS) ← largest

Case 2

$k \geq 0$

$f(n) = \Theta\left(n^{\log_b a} \lg^k n\right)$ → $\Theta(f(n))$ → equal

$T(n) = \Theta\left(n^{\log_b a} \log^{k+1} n\right)$ or $T(n) = f(n) \times \log n$

Case 3

$\epsilon > 0$

$f(n) = \Omega\left(n^{(\log_b a) + \epsilon}\right)$

$T(n) = \Theta(f(n))$ ← $f(n)$ → largest

1) $T(n) = 2T\left(\frac{n}{2}\right) + n$

$a=2 \quad b=2 \quad f(n)=n$

$w.s = n^{\log_b a} = n^{\log_2 2} = n$

$w.s = f(n) \rightarrow$ any

Case 2

$T(n) \; \Theta(n \times \lg n)$

$f(n) = \Theta(n)$

How?

Let's Prove

$f(n) = \Theta(n^{\log_b a} \log^{k} n)$

$T(n) = \Theta(n^{\log_b a} \log^{k+1} n$

$f(n) = \Theta(n^{\log_b a} \log^{k} n) \quad k \geq 0$

$f(n) = \Theta(n^{\log_2 2} \log^{0} n)$

$f(n) = \Theta(n^{1} (1)) = \Theta(n)$

$T(n) = \Theta(n^{\log_b a} \log^{k+1} n)$

$T(n) = \Theta(n^{(1)} \log^{0+1} n)$

$T(n) = \Theta(n \lg n)$

2) $T(n) = 2T\left(\frac{n}{2}\right) + \sqrt{n}$

$a=2 \quad b=2 \quad f(n)=\sqrt{n}$

$w.s = n^{\log_2 2} = n$

$f(n)$

$WS > f(n)$

Case 1

$f(n) = \Theta(n^{\log_b a - \epsilon}) \quad \epsilon > 0$

$T(n) = \Theta(n^{\log_b a})$

$f(n) \leq \Theta(n^{\log_2 2 - \epsilon})$

$f(n) \leq \Theta(n^{\log_2 2 - \epsilon})$

$\sqrt{n} \leq n^{\log_2 2 - \epsilon}$

$n^{1/2} \leq n^{1-\epsilon}$

$\epsilon = 0.5$

$T(n) = \Theta(n^{\log_b a})$

$T(n) = \Theta(n^{\log_2 2}) = \Theta(n)$

$f(n) = \Theta(n^{1-\epsilon})$

$\epsilon \geq 0.5$

3) $T(n) = 2T\left(\frac{n}{2}\right) + n^2$

$f(n) = n^2$, $a=2$ $b=2$

$WS = n^{\log_b a} = n$

$f(n) > WS$ Case 3

In Short $f(n) = \Theta(WS) = \Theta(n)$

$T(n) = \Theta(f(n)) = \Theta(n^2)$

But we use formula To Demonstrate or Prove

$f(n) = \Theta(n^{\log_b a + \epsilon})$

$T(n) = \Theta(f(n))$

$f(n) = \Theta(n^{\log_2 2 + \epsilon}) = \Theta(n^{1+\epsilon})$

$n^2 \geq \Theta(n^{1+\epsilon})$ $n^2 \geq n^{1+\epsilon}$ $1+\epsilon = 2$

$\epsilon = 1$

$f(n) = \Omega(n^{1+\epsilon})$ where $\epsilon \geq 1$

$T(n) = \Theta(f(n)) = \Theta(n^2)$

$f(n) = \Theta($

a) $= T(n) = 2T(n/4) + 1$

$a = 2 \quad b = 4 \quad f(n) = 1 \qquad W.S = n^{\log_b a} = n^{\log_4 2} = n^{\log_4 4^{1/2}} = n^{1/2}$

$f(n) = 1 \quad ; \quad WS = n^{1/2}$

$WS > f(n)$

$f(n) = \Theta(n^{(\log_b a) - \epsilon}) = \Theta(n^{1/2 - \epsilon})$

$1 = n^{1/2 - \epsilon}$

$n^0 = n^{1/2 - \epsilon}$

$0 = 1/2 - \epsilon$

$\epsilon = 1/2$ So

$$\boxed{\begin{array}{l} f(n) = \Theta(n^{1/2 - \epsilon}) \text{ where } \epsilon \geq 1/2 \\ T(n) = \Theta(n^{\log_b a}) = \Theta(n^{1/2}) \end{array}}$$

Ans

b)

$f(n) = \sqrt{n}$ $\quad$ $Ws = \sqrt{n}$

So

$\Theta = \sqrt{n} \lg n$

Rough work

Noted during lec

DAA - Master Theorem

- **Recursion** $T(n) = aT(n/b) + f(n)$

Constants $\begin{cases} a > 0 \\ b > 1 \end{cases}$

$f(n)$ +ve

1) $WS > f(n)$ deciding fact

2) $WS = f(n)$

3) $WS < f(n)$

watershed function $n^{\log_b a}$

$\Theta(n^2)$

$T(n) = 2T(n/2) + n^2$

$a = 2$ $\quad f(n) = n^2$
$b = 2$

$WS = n^{\log_b a} = n^{\log_2 2} = n$

$\Theta(n)$

$T(n) = 2T(n/2) + \sqrt{n}$

$W.S = n^{\log_b a} = n$

$f(n) = n^{0.5}$

$T(n) = 2T(n/2) + n$

$WS = n \qquad f(n) = n$

$\Theta(n \times \lg n)$

Case 1

$f(n) = O(n^{\log_b^a - \epsilon})$ $\epsilon > 0$

the $T(n) = (\Theta\, n^{\log_b^a})$

$f(n) \leq O(n^{\log_b^a - \epsilon})$

$\sqrt{n} \leq n^{\log_n^a - \epsilon}$

$\sqrt{n} \leq n^{1-\epsilon}$

Case 2

$f(n) = \Theta(n^{\log_b^a} \lg^k n)$ $K \geq 0$

$T(n) = \Theta\, n^{\log_b^a} \lg^{K+1})$

$n = n^{\log_2^2} \lg^0 n$

$n = n$

$T(n) = \Theta\, n^{\log_b^a} \lg^{K+1} n$

$T(n) = O(n \lg n)$

Case 3

ic $f(n) = \Omega(n^{\log_b a+\varepsilon})$ $\varepsilon > 0$

the $T(n) = (\Theta f(n))$

$n^2 = \Omega\, n^{1+\varepsilon}$

so $\boxed{\varepsilon = 1}$

$T(n) = (\Theta f(n))$

$T(n) = \Theta(n^2)$

1) $T(n) = 9T(n/3) + n$

$T(n) = T(2n/3) + n$

$T(n) = 3T(n/4) + n\lg n$

$\lg_3 9$ $\quad 3^{\boxed{2}}$ $\quad \frac{9}{3}$

1) $a = 9$ $b = 3$ $f(n) = n$ $\quad ws = n^{\lg_b a} = n^{\lg_3 9} = n^2$

$\Theta(n^2)$

$a = 1$ $b = \frac{3}{2}$ $f(n) = n$ $\quad ws = n^{\lg_{3/2} 1}$ $\quad n^{3/2}$

~~ws = n~~ $\cancel{\Theta(n^{3/2})}$ $\Theta(n)$

$a = 3$ $b = 4$ $f(n) = n\lg n$ $\quad ws\ n^{\lg_3 4} =$

$\Theta(n\lg n)$

$T(n) = T\left(\frac{2n}{3}\right) + 1$

$ws = n^{\lg^{0}_{3/2}} = n^0 = \underline{1}$ $\varepsilon = 0$

$\Theta(1)$

1) $f(n)$, a, b, ws, Θ, ε, K

1) $n \lg_2^{2^n}$ a if constant, So can't apply

2) $f(n) = 2^n$ → expression so can't apply

Can't Solve

$T(n) = \underline{2^n} T(n/2) + n$

$T(n) = 2\, T(n/4) + \underline{2^n}$

$T(n) = \underline{0.5}\, T(n/6) + n \lg n$

$T(n) = 2T(n/2) + \dfrac{\lg n}{n}$

$\frac{n}{\lg n}$ also

Only Solve By Extended Master Theorem

$T(n) = T(n/9) \underline{- n}$